Hector Cordero

hector.corderod@gmail.com | linkedin.com/in/hectorhcordero | github.com/hector1128 | hectorcordero.com

Education

University of Central Florida (B.S. in Computer Science, Minor in Mathematics) – 3.8 GPA — May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Software Development, Database Systems, Operating Systems, Network Optimization

Clubs/Organizations: ColorStack, Knight Hacks, SPHE, F1 AI Racer

Experience

CarMax — June 2026 – Aug. 2026

Software Engineer Intern – Online Appraisals Team — *Richmond, VA*

- Shipped the photo upload platform for vehicles ineligible for an instant offer in **React** with a **.NET** backend (**1M+ users**), increasing offer completion on the ineligible segment from **35% to 62%**.
- Engineered a reusable **TypeScript** image-processing tool (HEIC-to-JPEG conversion and client-side compression) for a 9–12 photo capture flow, improving upload speeds while maintaining high-quality images for buyers.
- Created an isolated QA environment for mobile testing by creating IaC templates (**YAML/Bicep**), configuring an **Azure App Service** and building an automated CI/CD pipeline in **Azure DevOps**.
- Executed the end-to-end testing plan with **Playwright** to eliminate regressions, then analyzed real user sessions via **FullStory** and **Adobe Analytics** to iterate on the user experience, cutting completion times by **35%**.

Knight Hacks — Sep. 2025 – Present

Software Engineer — *Orlando, FL*

- Led core engineering for a hackathon's form-builder platform, distributing the data collection through a pipeline to securely ingest and process records for a **1,800+** member student organization.
- Containerized production infrastructure using **Docker**, self-hosting multi-app deployments to guarantee zero-downtime scaling for **1,200+** concurrent hackers.
- Secured routes for using **Discord OAuth** and **NextAuth.js** for role-based access control, preventing cheating and reducing admin workload by **30%**.
- Optimized data distribution across teams by implementing a real-time Kanban board utilizing **tRPC** and state hydration, improving internal communication across teams and cutting work time by **22%.**

Krystal Jewels — Sep. 2025 – Present

Co-Founder - Software Engineer — *Plantation, FL*

- Scaled and maintain a full-stack jewelry e-commerce platform serving **1,000+** customers, owning end-to-end development across storefront, inventory, and order management, generating over **$15K in gross revenue.**
- Deployed **agentic AI workflows** (Claude Code) to automate marketing content generation, inventory management, analytics tracking, and financial analysis, replacing manual processes across the team.
- Built an **AI-powered PDF invoice parser** that extracts and stores structured product data into **Supabase/PostgreSQL**, cutting manual inventory entry time by **65%**.
- Generated **$5K** in new sales by designing end-to-end **TikTok** marketing campaigns and managing 5+ creator partnerships from outreach to contract.

Projects

Trackr | *Azure Event Hub, Azure Functions, Azure Cosmos DB, React.js, Azure Maps* — Aug. 2026

- **3rd** out of **53** teams at CarMax's Innovation Garage hackathon, where I built a real-time fleet-tracking platform for vehicle transfers.
- Ingested live GPS and telemetry data from **Samsara** and **Recovr** via a **Azure Event Hub**, processing records through **Azure Functions** into a live **Cosmos DB** truck data store.
- Built a live-ops dashboard with an interactive map, filterable alert panel, and per-truck static/live detail views, giving logistics admins and associates real-time visibility into **150+** trucks without manual dispatch lookups.
- Built an internal debug console for direct queries to **Cosmos DB** containers + live GPS event streaming for **40%** faster iteration.

InvestEd | *Next.js, TypeScript, Gemini 2.5 Flash, Zustand, shadcn/ui, Vercel, Tailwind* — Oct. 2025

- **2nd** out of **68** projects at the Knight Hacks 2025 OneEthos AI for Financial Empowerment Challenge.
- Integrated **Gemini 2.5 Flash** chat bot to deliver live financial guidance at **200ms** average latency.
- Built an automated HYSA finder ranking real-time APY data across **50+** banks via a custom scoring algorithm.

MOVE | *Next.js, TypeScript, MediaPipe Pose, TensorFlow, WebGPU, WebGL, Tailwind* — Sep. 2025

- Shipped a **computer vision** exercise app using **TensorFlow + WebGPU** acceleration, rendering **33 MediaPipe Pose** keypoints per frame across custom **HTML5** Canvas layers.
- Reduced false-positive collision detections by **85%** with an **EMA** filter and sustained **60 FPS** on low-end hardware through render throttling and progressive difficulty scaling.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, C#, C++, SQL, HTML/CSS, YAML, Bicep

Technologies: Next.js, React, Node.js, Azure, AWS, PostgreSQL, Supabase, Prisma, Firebase, Docker, TensorFlow, Figma, MediaPipe, WebGPU, WebGL, NextAuth.js, Zustand, tRPC, Vercel, Kubernetes, Cypress, Playwright

Concepts: Full-Stack Development, AI Integration, REST APIs, Prompt Engineering, UI/UX Design, Computer Vision, CI/CD Pipelines, Distributed Systems, System Design